VTPEH 6105

Lecture 01

Course Overview, Intro to R & Descriptive Statistics Refresher

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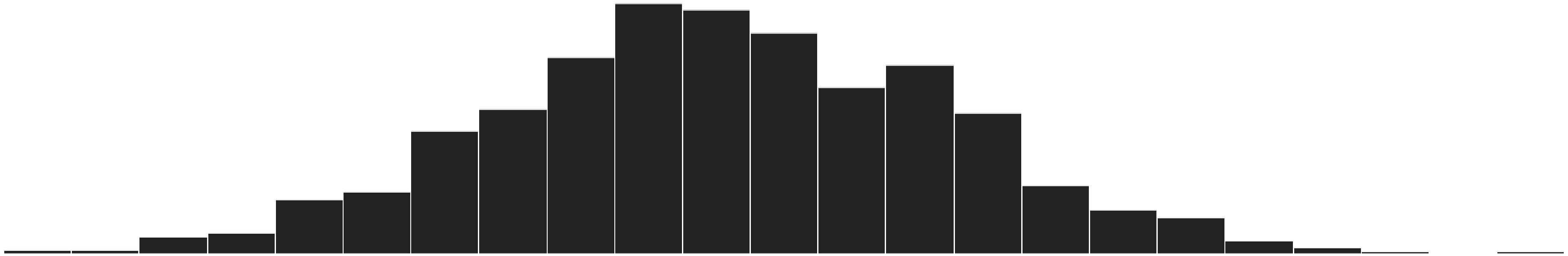


Course Overview

About the Teaching Team

- Instructor

Amandine Gamble

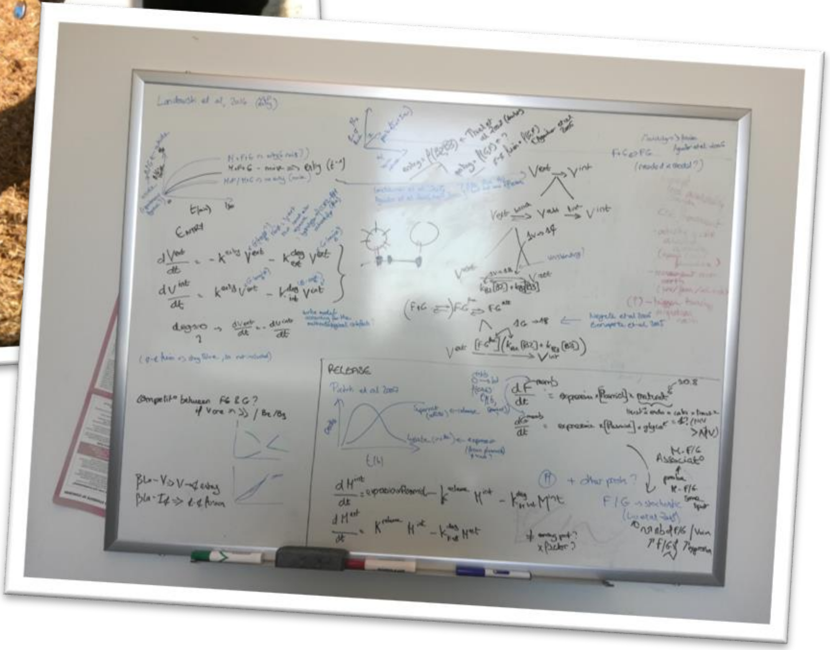
Amanda + Marilyn

Infectious disease dynamics in wildlife and at the human-wildlife interface

Assistant Professor in the Department of Public and Ecosystem Health

Learn more about my research at
amandinegamble.github.io

From veterinary medicine...



... to mathematical modelling

About the Teaching Team

- Course Assistants

  Jordan Deskins

  Anika Jessup

  Mainul Islam

  Princal Patel

  Madison Wolfinger

  Daniel Addo (VTPEH 6108)

  Harsh Sonar (VTPEH 6108)

  Dan Lin / Therese Martin Cheteu Wabo (VTPEH 6108)

Contact details on Canvas

Course Goals

- Council on Education for Public Health (CEPH) Competencies
  - 3.3. Analyze quantitative (and qualitative) data using biostatistics, informatics, computer-based programming, and software, as appropriate.
  - 3.4. Interpret data analysis to guide public health research, policy, or practice.

VTPEH 6105: Biostatistics for Health Sciences

- Focus is on **statistical concepts and practices: statistical literacy and capability**
  - **Assess and interpret evidence** produced by others (scientific literature, project reports…)
  - **Produce evidence yourself**
    - **Analyze data**, and everything before (design, collect…) and after (interpret, communicate…)
    - Get **R basics** as a **foundation** for data collection and management, collaborations…

Assignments

- Path towards the statistical literacy and capability
  - **Attendance and Participation in Class Activities** *5% of final grade*
    - Help each other with in-class practice and homework
  - **Weekly Knowledge Checks** *25% of final grade*
    - Quiz on Canvas, designed to assess your understanding of the key concepts ahead of next class
  - **Weekly Homework: Practice and Peer-Review** *45% of final grade*
    - Data analysis, coding, and/or peer-review
  - **Weekly Homework: Guiding Quizzes** *5% of final grade*
    - Quiz on Canvas, with guiding questions to help you with the homework
    - Excused by default; required if previous assignments suggest you are struggling
  - **Final Exam: Peer-Review** *20% of final grade*
    - In-class peer-review of a publication
    - Aimed to assess your **understanding of the key concepts** and **critical thinking**

Course Structure

- Proficiency through **practice**

  Each week:
  - **Lecture** focused on **statistical concepts and approaches** (~40 min)
  - **Hands-on workshop** (~40 min)
  - **Knowledge check** (~5 min) – First attempt; unlimited retake opportunities
  - **Collective working session** (~40 min) – Perfect moment to ask questions, work on your homework…

- Feel free to bring your lunch

Provisional Schedule

Lecture	Lecture	Workshop
#01 01/23	Descriptive statistics refresher	*R how-to*: getting started and exploring data
#02 01/30	Inference: probabilities and uncertainty	*R how-to*: complex coding with AI and practice with data wrangling
#03 02/06	Introduction to hypothesis testing and p-values	*Peer review how-to*: introduction and focus on the statistical analyses
#04 02/13	Comparisons of means across two groups	*R how-to*: continuous data and hypothesis testing
#05 02/20	Comparisons of means across several groups and analyses of variances	*R how-to*: continuous data and hypothesis testing
#06 02/27	Comparisons of prevalences across groups	*R how-to*: categorical data and hypothesis testing
#07 03/06	Comparing what is comparable: adjustment and standardization	*Excel and R how-to*: population adjustment and standardization
#08 03/13	Linear and logistic regressions	*Excel and R how-to*: regression models
#09 03/20	Studying more complex associations: multivariate models	*R how-to*: multivariate models
#10 03/27	Beyond p-values: unmasking statistical fallacies	*Peer review how-to*: looking beyond the statistics
04/03	Spring break	
#11 04/10	Other regression models: survival, additive, mixed…	*Peer review how-to*: working around new methods
#12 04/17	Analyzing questionnaire data	*R how-to*: working with questionnaire data
#13 04/24	Introduction to high-dimensional data analysis and machine learning	Mini journal club
#14 05/01	Introduction to mechanistic modeling	*R how-to*: tailoring existing models
- 05/08	*Study period*	
- 05/15	*Exams*	Exam

Additional Support and Practice Opportunities

- Everyone is enrolled in **VTPEH 6108: Public Health Data Lab** by default:
  - **Weekly in-class support**
  - Perfect moment to **work on your homework**
  - Or to **work on your APEx data** if you have any
  - Or to ask any question you have related to statistical analysis, R coding, or **anything else related to quantitative data**

- Want to go further with coding? Consider joining **VTPEH 6270: Data Analysis with R** (instead of VTPEH 6108: Public Health Data Lab)

VTPEH 6270: Data Analysis with R – Enrollment

- Passing the 1st assignment (Check Point #01) is required to assess basic proficiency, hence enroll
  - Due on Monday January 26, 10 AM
  - Grade posted on Friday January 30
  - Enrollment possible until Tuesday February 03

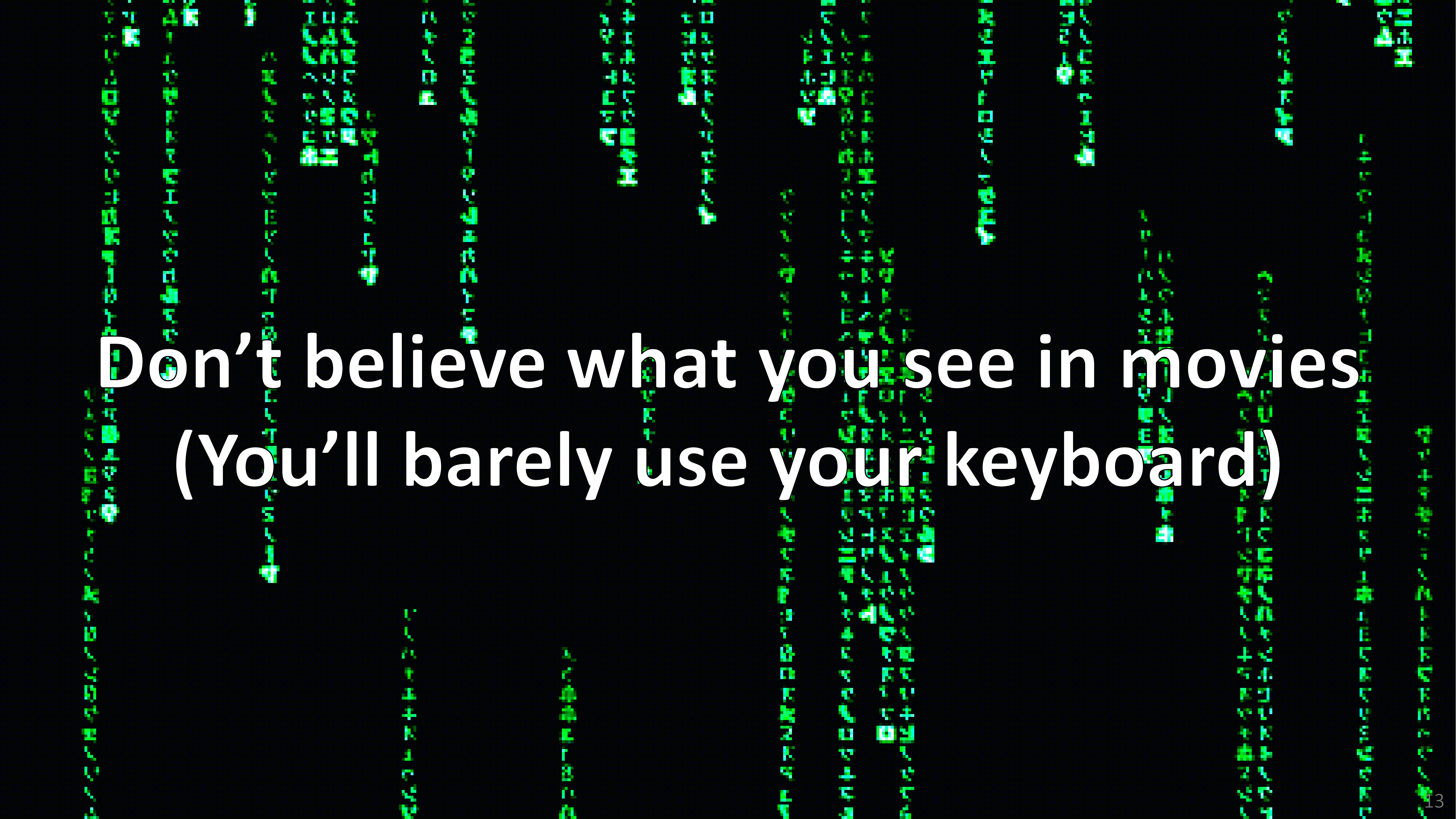
- Please fill this form to be pre-enrolled and access the Canvas page

tinyurl.com/MPH-VTPEH6270-Pre

Intro to R

Why R in Public & Ecosystem Health?

- Open source (free and transparent)
- Increasingly used in public health (e.g., CDC)
- Save workflows in scripts, from data formatting to report generation
  - Automation → power and efficiency of science
  - Reproducibility → robustness of science
  - Transparency → access to science
- Advanced analytical techniques
  - Epidemiological modeling
  - Spatial analyses
  - Web scraping
  - …



Don’t believe what you see in movies

(You’ll barely use your keyboard)


Google
how to pivot table r
AI Mode

Getting Started…

R Bootcamp (available on Canvas)

Generating Reports with Rmarkdown

.Rmd file
File -> New File -> R markdown file...

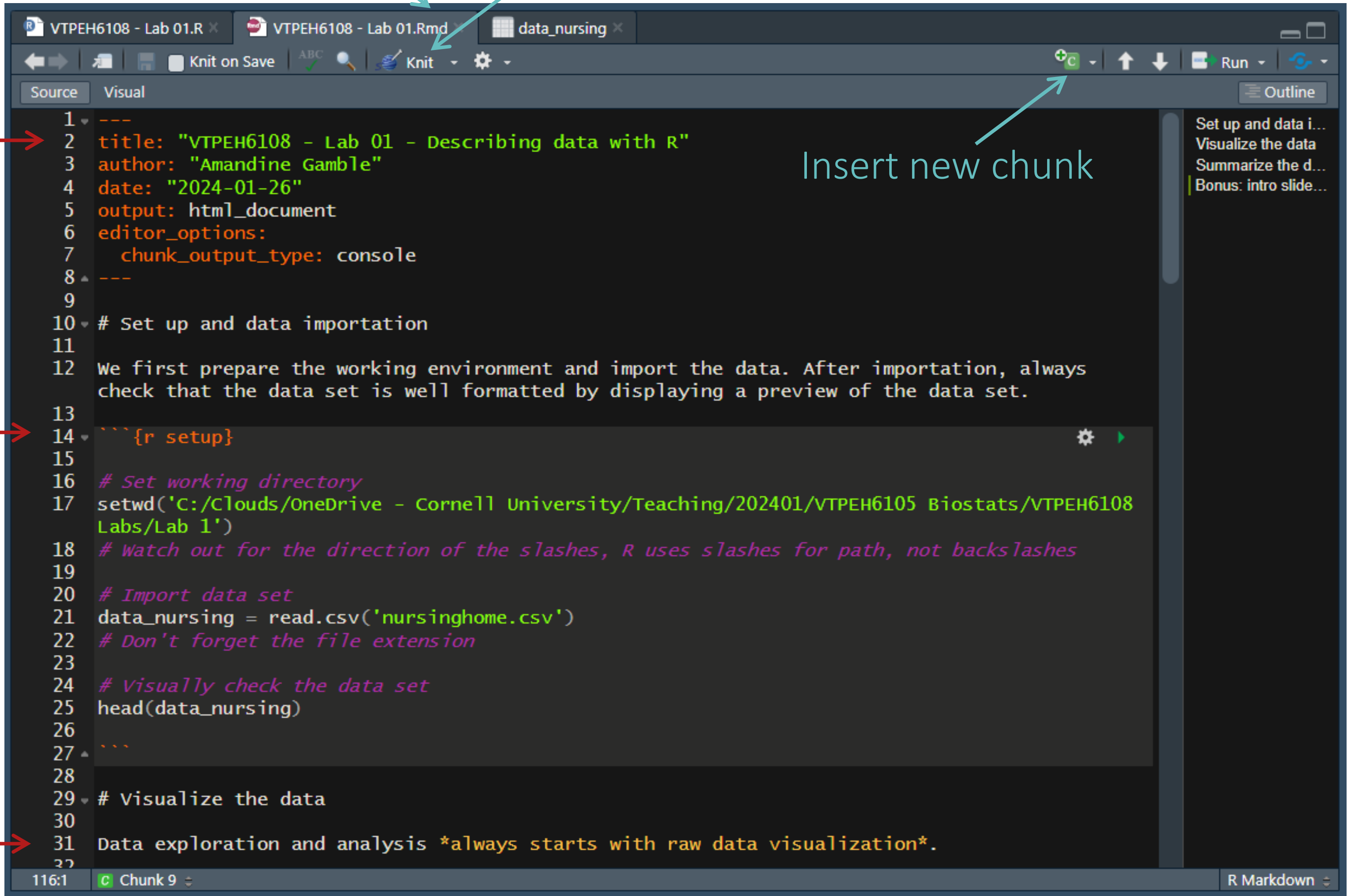
Generate final document

Insert new chunk

Document set up

Code chunk

Literate text



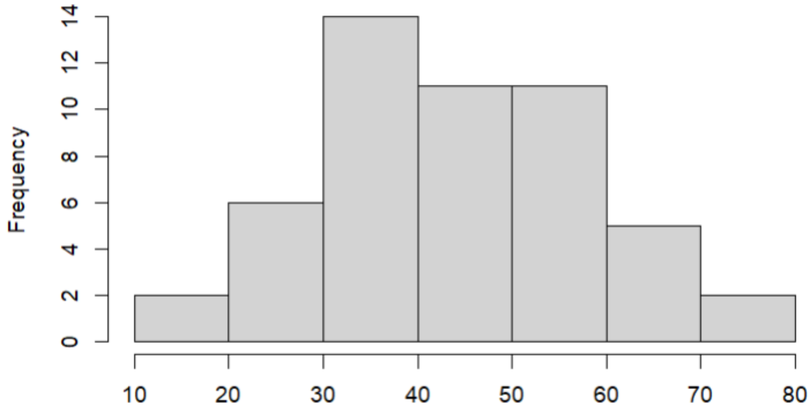
Let’s Try Together…

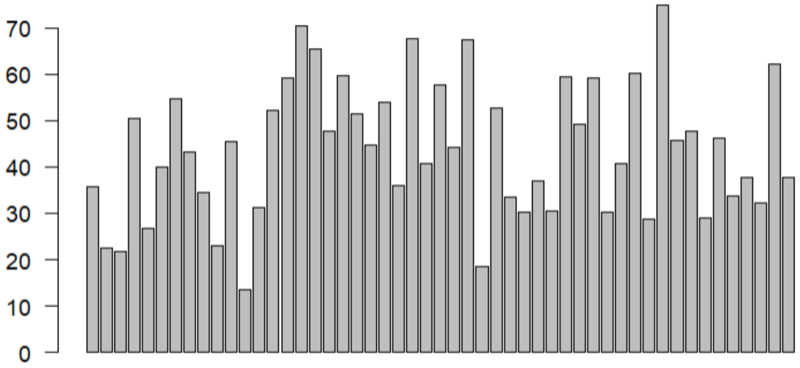
Homework #01 (available on Canvas)

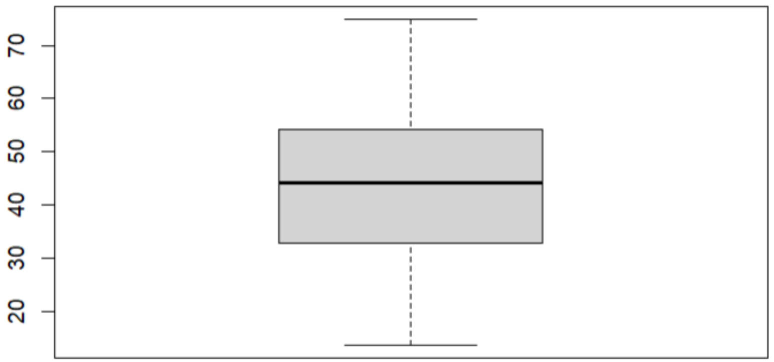
Descriptive Statistics Refresher

Describing Data… **Always Visualize the <u>Raw</u> Data First**

- Visualizing data
  - Histogram with `hist()`
    Distribution
  - Barplot with `barplot()`
    Individual point
  - Boxplot with `boxplot()`
    Summary statistics







Which one is more informative?

On the Parsimonious Use of Data Summary

Information

Histogram

+++

Jittered scatter plot

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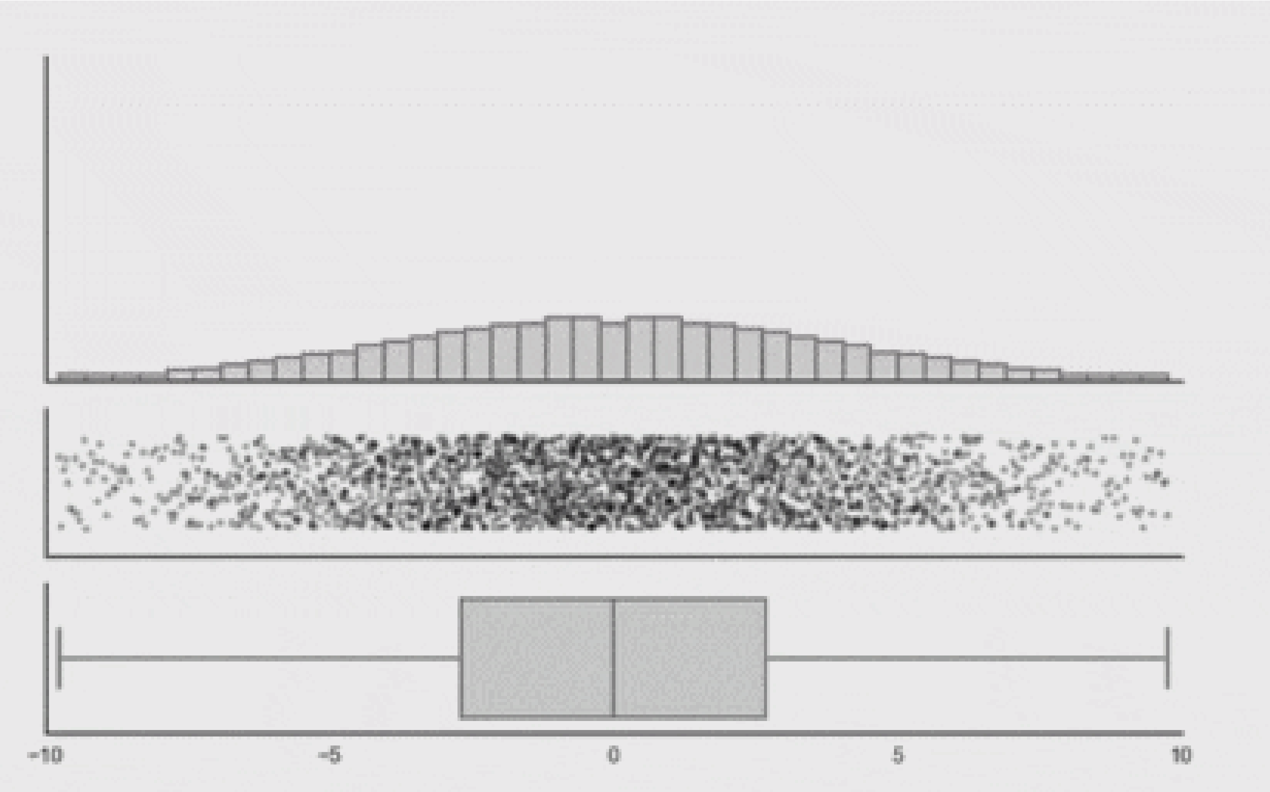
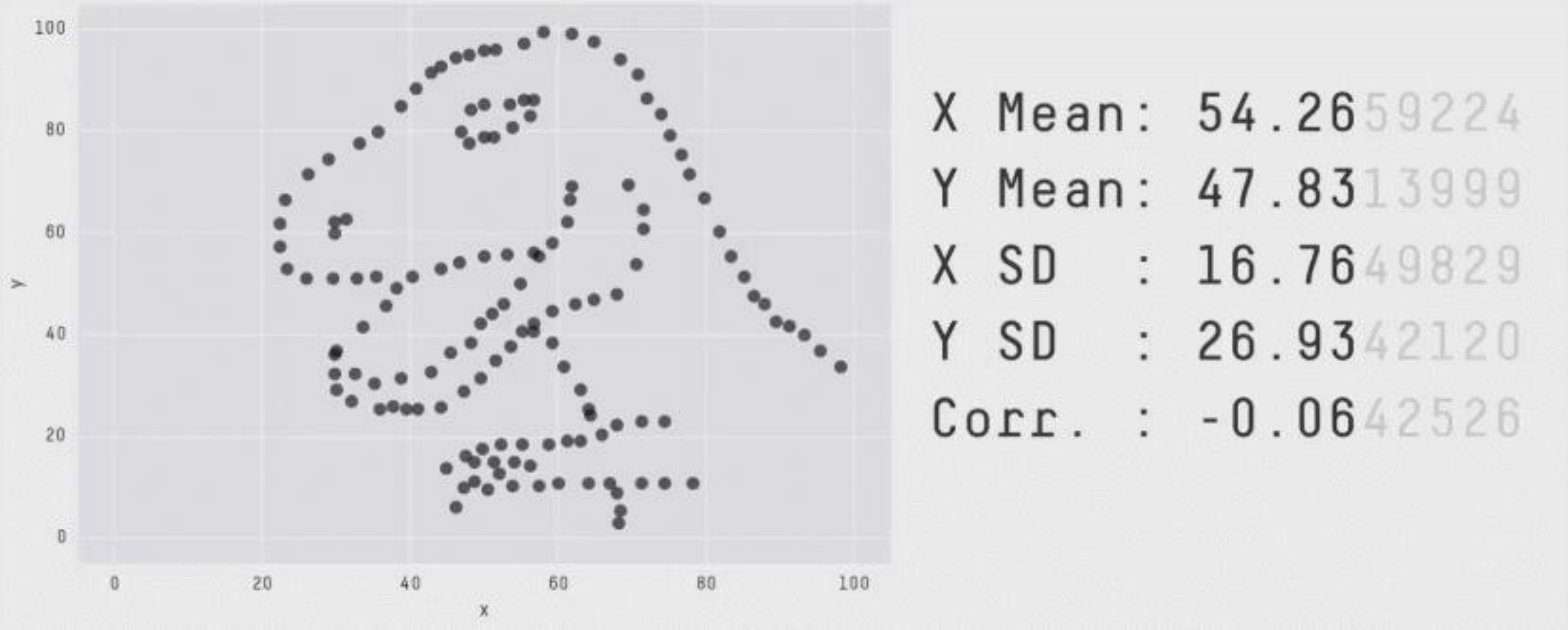
Boxplot

Summary statistics

-



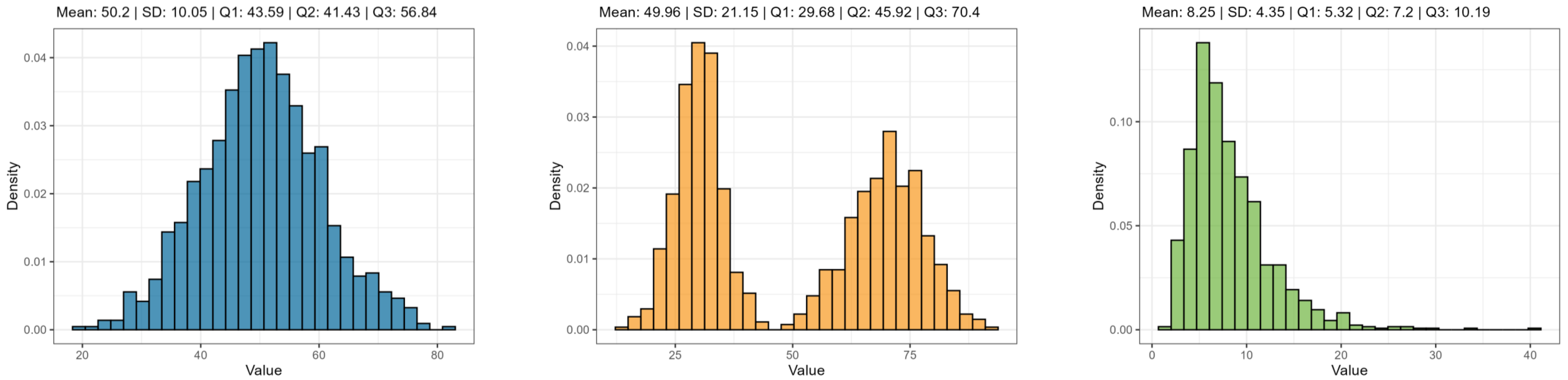
On the Parsimonious Use of Data Summary



They are useful, only when accompanied by the raw data (histogram or scatter plot)

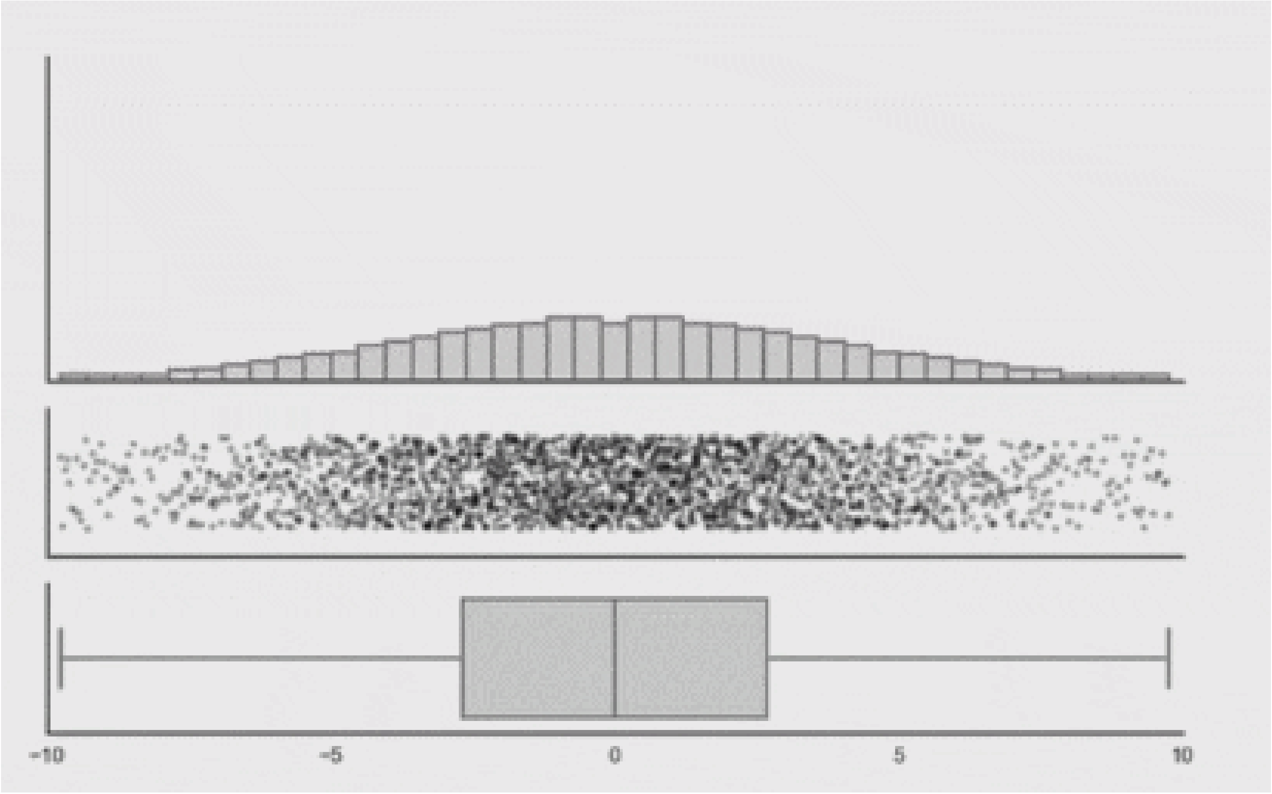
The DataSaurus, or "same stat, different graph", from Autodesk

On the Appropriate Use of Data Summary



- Think about how well the picked summary statistics describes the distribution of the data (symmetrical vs skewed, uni- vs multimodal…)

To Conclude...



Visualize

Google (or AI)

Before Next Time (Friday January 30)

- Carefully read the syllabus if you have not already
- Practice: Homework #01 due by **Thursday 5 PM**
- Check you are ready for next class: Knowledge Check #01 due by **Friday 11 AM**

Need Help?

- Go to VTPEH 6108: Public Health Data Lab
- Go to office hours (to be announced)
- Go over R basics once again
    - Cornell Statistical Consulting Unit: The Essentials of R for Statistical Analysis Day 1: Getting Started with R